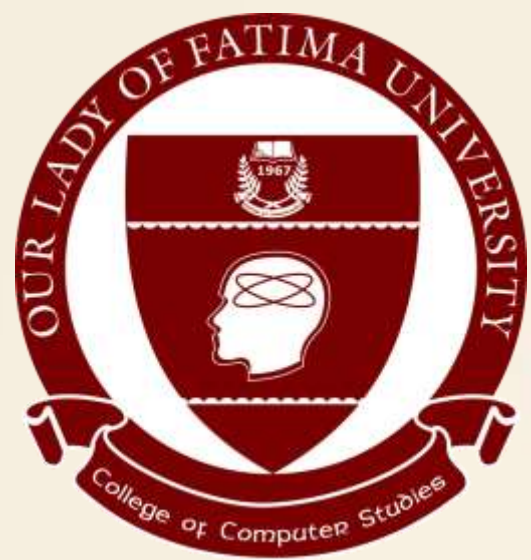




Module 13-15

Current Trends in Computing

ITCL112 - INTRODUCTION TO COMPUTING W/ LAB

INTRODUCTION



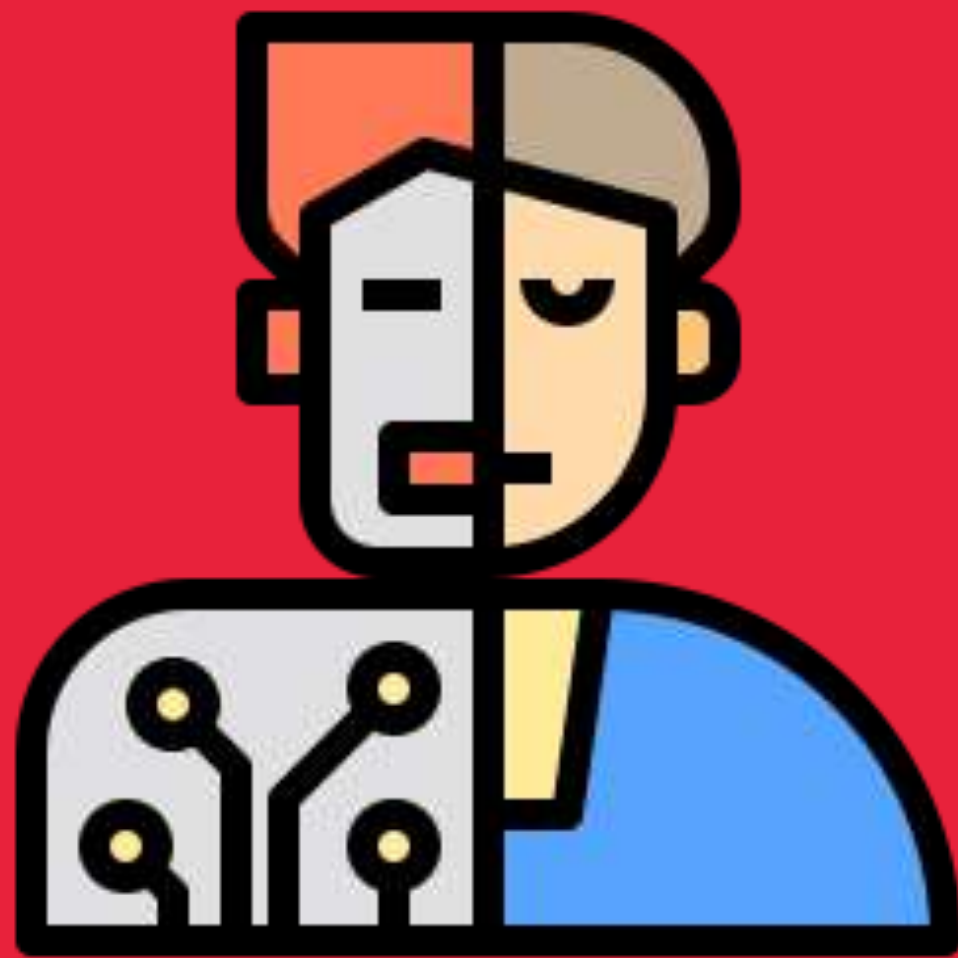
This module bridges the gap between global innovations and the Philippine technological landscape, offering a deep dive into the mechanics of tools reshaping our future in 2025. Beyond just defining breakthroughs like Agentic AI and Satellite Connectivity, this module analyzes how government initiatives—such as DOST’s GeoRiskPH and the DICT’s Digital Cities roadmap—are applying these technologies to solve local challenges in disaster management and governance. Ultimately, students will gain a strategic understanding of how to leverage these trends to transition from traditional roles into high-value career paths like Clinical Coding, Cloud Engineering, and Cybersecurity within the country's evolving digital economy.

MODULE 10

After this lesson, students should be able to:

- 1. Understand how the top technologies work and differentiate them from older versions.**
- 2. Analyze specific Philippine government projects (DOST/DICT) that utilize these technologies.**
- 3. Know the detailed breakdown of the specific new jobs mentioned in the IBPAP Roadmap 2028.**

Emerging Technologies



Agentic AI (vs. Generative AI) - Most people know ChatGPT, which is Generative AI. You ask a question, it writes an answer. It is passive. An "Agent" is proactive. It has "limbs" (access to other software) and "reasoning" (ability to plan steps).

How it works:

- **Generative:** You ask, "Plan a travel itinerary to Boracay." It gives you a list.
- **Agentic:** You say, "Book me a trip to Boracay under ₱15k." The AI Agent autonomously:
 1. Checks flight prices on PAL/Cebu Pacific.
 2. Checks Agoda for hotels.
 3. Executes the booking using your credit card API.
 4. Adds the dates to your Google Calendar.

Philippine Impact: BPO companies are shifting from hiring "Customer Support Agents" to "AI Agent Managers"—humans who supervise 50+ AI agents handling refunds and bookings simultaneously.

Emerging Technologies



Satellite-to-Phone Connectivity (Starlink/SpaceX) - "Satellite Phones" were bulky, expensive brick-phones used by soldiers or sailors. Your normal smartphone connects directly to a Low Earth Orbit (LEO) satellite without needing a cell tower.

How it works:

- LEO satellites orbit very close to Earth (550km vs. the old 35,000km). Because they are closer, regular phone antennas can pick up their signal.

Why it matters in the PH:

- **Disaster Proofing:** When a typhoon topples cell towers in Bicol or Samar, this tech allows people to send SMS/SOS signals directly to the sky, bypassing damaged ground infrastructure.
- **Konektadong Pinoy Act:** A push to allow more satellite internet providers into the PH to break local duopolies.

Real-World Applications



HazardHunterPH
GEORISKPH

Disaster Risk Reduction: GeoRiskPH & Project VOLCAN

- ✓ A multi-agency initiative led by DOST-PHIVOLCS.
- ✓ It is a central database that combines data from satellites, fault finders, and historical flood records.

Real Application (HazardHunterPH):

- Before a developer builds a condo or a mayor approves a subdivision, they must use HazardHunterPH.
- They pin the location on the map, and the system generates a "Hazard Assessment Report" in seconds, detailing if the land sits on a fault line, is prone to liquefaction, or is in a flood zone.

Project VOLCAN: specifically monitors volcanic unrest (like Taal and Mayon) using AI to detect subtle ground deformation patterns that humans might miss, predicting eruptions earlier.

Real-World Applications



The "Digital Cities 2025" Program

- ✓ The DICT identified 25 cities aimed to be the next "IT Hubs" to decongest Metro Manila.

Why this matters to you: If you live near these cities, you don't need to move to Manila for a high-paying tech job.

The List (Key Examples):

- Luzon: Balanga, Batangas City, Cabanatuan, Iriga, Laoag, San Fernando (La Union & Pampanga), Tarlac, Tuguegarao.
- Visayas: Roxas City, Tacloban, Tagbilaran.
- Mindanao: General Santos, Iligan, Zamboanga.

What makes them "Digital"? These cities are receiving priority budget for fiber optic internet and "Technology Business Incubators" (TBIs) for startups.

IT Careers in the PH



Health Information Management (HIM)

The Job: Clinical Coder / Medical Coder.

Why it's huge: The Philippines is the #1 destination for US healthcare BPO. Nurses who don't want to work bedside are moving here.

What you actually do:

- ✓ You review a doctor's notes (e.g., "Patient has high fever, joint pain, rash").
- ✓ You translate this into universal alphanumeric codes (ICD-10). Example: You convert "Dengue Hemorrhagic Fever" into code A91.
- ✓ Why? US Insurance companies and computers cannot read doctor's handwriting; they only read code A91 to process the payment.

Salary Potential: High, because it requires certification (like AAPC) and medical terminologies.

IT Careers in the PH



Cybersecurity: SOC Analyst

The Job: Security Operations Center (SOC) Analyst.

What you actually do:

- ✓ You sit in a "War Room" (or remote dashboard) monitoring network traffic.
- ✓ The Task: You don't "hack" people. You watch for "Anomalies." Example: "Why is an employee's account trying to download 50GB of files at 3:00 AM on a Sunday?"
- ✓ You investigate and block the threat (using AI tools) before data is stolen.

Requirement: DICT is heavily promoting "Micro-credentials" (short certifications like CompTIA Security+) over 4-year degrees for this specific role.

IT Careers in the PH



Cloud Engineer (DevOps)

The Job: Managing the "invisible computers" that run apps.

What you actually do:

- ✓ Instead of buying a physical server worth ₱500k, you "rent" computing power from Amazon (AWS) or Google (GCP).
- ✓ You write scripts that automatically "spin up" more servers when traffic is high (like during a 11.11 Sale on Lazada) and "turn them off" when traffic is low to save money.

IT Careers in the PH



Other Hot Jobs

- ✓ **UI/UX Designer (User Interface / User Experience)** - You don't just make apps "look pretty"; you design how they feel. You study human behavior to ensure a user doesn't get frustrated trying to find the "Checkout" button. (Figma, Adobe XD, Sketch.)
- ✓ **IT Project Manager / Scrum Master** - You don't write code; you unblock the people who do. You run the daily meetings ("stand-ups"), manage timelines, and ensure the team hits the deadline without burning out.
- ✓ **ESG Data Analyst (Environmental, Social, and Governance)** - A very new role. You use data tools to track a company's carbon footprint and sustainability metrics. (Salary: Can range from ₱40k - ₱80k for specialized roles.)
- ✓ **Data Privacy Officer (DPO)** - You ensure the company follows the Data Privacy Act of 2012. You don't need to be a lawyer, but you need to understand how data flows in IT systems to prevent leaks.
- ✓ **Automation Specialist** - You connect different apps to talk to each other.
- ✓ **Low-Code / No-Code Developer** - Building websites and apps without writing traditional code. You use drag-and-drop platforms to build MVPs (Minimum Viable Products) for startups in weeks instead of months.